HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed assembly, integration, and test procedures for flight and engineering units, ensuring adherence to strict quality expectations and customer audit requirements.
- Assembled and integrated complex mechanical, electronic, and embedded computer sub-assemblies into complete avionics units, meticulously managing component handling in **cleanroom** and **ESD-controlled environments**.
- Configured and verified embedded computer systems, including firmware loads and software interfaces, during unit-level integration and testing phases, utilizing laboratory test equipment for validation.
- Investigated and resolved faults and test anomalies, actively contributing to root-cause analysis and non-conformance reporting to drive corrective actions and process improvements.
- Maintained detailed as-built configuration records and prepared comprehensive delivery documentation, including end-item data packages, for flight hardware.
- Collaborated with engineering teams to provide feedback on designs for ease of assembly, integration, and test, directly supporting new product introduction activities.
- Containerized the feature-pipeline services with Docker and deployed them to a Kubernetes cluster (EKS), wiring up Helm chart configuration and basic resource limits for predictable rollout.

SKILLS

Avionics & Mechatronics – Mechatronics Engineer, Spacecraft Avionics, Electronic Units, Embedded Computer Systems, Flight Hardware, Engineering Units, Qualification Models, Ground Support Equipment, Vibration Testing, Shock Testing, EMC Testing, Thermal Testing, Thermal Vacuum Testing, Cleanroom Standards, ESD Control, COTS Electronics, Space Hardware Standards, Workmanship Standards, IPC Standards

Engineering Tools & Software – Solidworks, Fusion, Autodesk, LabVIEW, Python, R, Java, Spring Boot, SQL, PostgreSQL, MySQL, Oracle, Splunk, Dynatrace, Jira, Confluence, Git, Jenkins, Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudWatch)

Programming Languages – Python, Java, SQL, R, English

Methodologies & Practices – Agile, Scrum, Process Driven, Hands-on, Resourceful, Communicative, Collaborative, Attention to Detail, DFM/DFT, Root Cause Analysis, Non-Conformance Handling, Change Management, Quality Management

Data & Analytics (MSc) – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, Data Modelling, Data Warehousing, Statistical Analysis, Regression, Hypothesis Testing, Exploratory Data Analysis, A/B Testing, Time-Series Analysis, Machine Learning Fundamentals, Big Data Concepts (Hadoop/Spark)

PROJECTS

Spacecraft Avionics Unit Integration Framework

- Developed a standardized framework for integrating and testing spacecraft avionics units, incorporating best practices for assembly, harness management, and functional verification.
- Designed and built custom test jigs and fixtures to facilitate repeatable assembly and testing of electronic sub-assemblies, reducing setup time by approximately 30%.
- Documented comprehensive test procedures and work instructions for flight hardware, ensuring compliance with space industry quality standards and customer requirements.

Embedded Systems Firmware Verification Suite

- Engineered a Python-based test suite to automate the bring-up, configuration, and verification of embedded computer systems for avionics applications.
- Integrated various laboratory test equipment and ground support software through Python scripts, enabling automated functional and performance testing of firmware loads.

- Implemented fault injection and anomaly detection mechanisms within the test suite to aid in root-cause analysis of embedded system failures.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)